> *"Just as a ball rolls downhill, a positive charge naturally moves from regions of high potential to regions of low potential — guided by the invisible landscape of the electric field."*
>
> *— Principle of Electric Potential*

WHY IT MATTERS

Electric potential is the concept that unifies all of electricity. Every battery, power outlet, and electronic device operates on voltage — which is simply electric potential difference. Without this concept, modern technology would not exist.

EVERYDAY CONNECTION

The "voltage" printed on a battery label represents the electric potential difference between its terminals. A 1.5 V battery means there is 1.5 joules of energy available for every coulomb of charge that moves through an external circuit.

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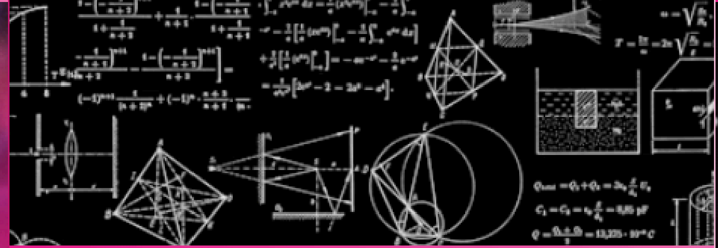

CHAPTER 3 — CORE CONCEPTS

Electric Potential Energy & Electric Potential

When a charged particle is placed in an electric field, the field exerts a force on it — and like any force over a distance, this constitutes **work**. The energy associated with a charge's position within an electric field is called **electric potential energy**, and the energy per unit charge at any point is called the **electric potential**.

Electric Potential Formula

$$V = PE / q$$

Potential (V) = Potential Energy (J) ÷ Charge (C)

V
Electric Potential
Unit: Volts (V) = J/C

PE
Potential Energy
Unit: Joules (J)

q
Electric Charge
Unit: Coulombs (C)

W
Work Done
$W = \Delta PE = -qE\Delta x$

Point charge potential: $V = kq / r$

Multiple charges (superposition): $V_{total} = V_1 + V_2 + V_3 + \dots$

Coulomb's constant: k = 8.99 × 10&sup9; N·m²/C²

COLLEGE PHYSICS 2 · UNIVERSITY OF MINDANAO

ELECTRIC POTENTIAL ENERGY

Understanding **Work, Electric Potential, Voltage** and Potential Due to Point Charges

Instructor: **Elmer E. Galang** · PHYS 102 · Week 1–3

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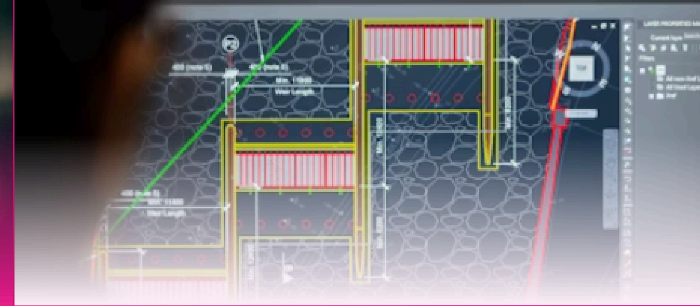


WORK & POTENTIAL ENERGY

Work Done by the Electric Field

When a charged particle moves within an electric field, work is done on it. This work is directly related to the **change in electric potential energy**. The relationship is expressed as $W = \Delta PE$, which is the work-energy theorem applied to electric charges.

- **Change in Potential Energy**
  $\Delta PE = -W_{ab} = -qE_x\Delta x$ · Work done against the electric field is stored as potential energy.
- **Direction of Movement**
  Moving a positive charge against the field increases PE. Moving it in the field's direction decreases PE — similar to gravity.
- **Conservative Force**
  The electric force is conservative — the work done depends only on start and end positions, not the path taken between them.

Sample Problem

A proton released at x = −0.020 m in a field of **1.50 × 10³ N/C** reaching x = 0.050 m.

ΔPE = −(1.60×10&sup9;)(1500)(0.070)
ΔPE = **−1.68 × 10&sup7; J**

ELECTRIC POTENTIAL & VOLTAGE

Voltage — Energy Per Unit Charge

Rather than measuring the total energy of a specific charge, it is more useful to express electric conditions in terms of **energy per unit charge**. This quantity is called **electric potential** — commonly known as **voltage**.

$$V = PE / q$$

Electric Potential = Potential Energy ÷ Charge

1 Volt = 1 Joule per Coulomb (J/C)

- **Voltage is a Scalar**
  Unlike electric field (a vector), electric potential is a scalar quantity — it has magnitude only, no direction.
- **Superposition Principle**
  The total potential from multiple charges is the algebraic sum of individual potentials: $V_{total} = V_1 + V_2 + \ldots$
- **Zero Reference Point**
  Electric potential is defined as zero at an infinite distance from the source charge, by convention.

Sample Problem

5.00 μC charge at origin, −2.00 μC charge at (3.00, 0) m. Find V at P(0, 4.00) m.

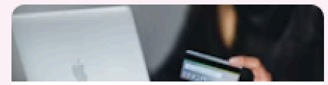
V_1 = (8.99×10&sup9;)(5×10&sup6;)/4 = **1.12×10&sup4; V**
V_2 = (8.99×10&sup9;)(−2×10&sup6;)/5 = **−0.36×10&sup4; V**
$V_P = V_1 + V_2$ = **7.6×10³ V**

SUMMARY & APPLICATIONS

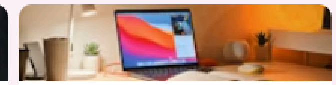
Key Concepts at a Glance

CONCEPT	FORMULA	UNIT
Potential Energy	ΔPE = −qEΔx	Joules (J)
Electric Potential	V = PE / q	Volts (V)
Point Charge V	V = kq / r	Volts (V)
Multiple Charges	V = Σ kq / r	Volts (V)

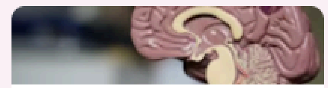
REAL-WORLD APPLICATIONS



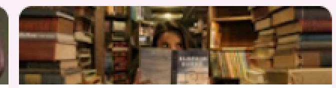
Batteries
Voltage rating shows potential difference between terminals.



Charging Devices
Chargers use potential difference to push charge into batteries.



Medical Devices
EKG and MRI machines rely on electric potential differences.



Power Grids
High-voltage lines transmit energy over long distances efficiently.

KEY TAKEAWAY

Electric potential energy describes the energy stored in a charge's position within a field, while electric potential (voltage) expresses that energy per unit charge. These concepts are the foundation of all electrical engineering, from household circuits to advanced medical technology.